blue/red: my attempt
black: proper answer

Upper Division Tutoring Program Topic Review

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Math 110 Midterm Exam Review

Basic Definitions, Theorems, and Reference

Definition 1 (Conditions for a subspace)**.** A subset U of V is a *subspace* of V if and only if U satisfies the following three conditions.

- Additive Identity ($0 \in U$)
- Closed Under Addition ($u, w \in U \implies u + w \in U$)
- Closed Under Scalar Multiplication ($a \in \mathbb{F}, u \in U \implies au \in U$)

Definition 2 (Linear Independence)**.** A list of vectors $v_1, v_2, \ldots, v_m$ in V is called *linearly independent* if the only choice of scalars $a_1, a_2, \ldots, a_m \in \mathbb{F}$ that makes

$$a_1v_1 + a_2v_2 + \cdots + a_mv_m = 0$$

is $a_1 = a_2 = \cdots = a_m = 0$. The empty list is also declared to be linearly independent.

Definition 3 (Basis and Dimension)**.** Let V be a vector space. A *basis* of V is a linearly independent list of vectors $\{v_1, \ldots, v_n\}$ that spans the space, i.e. any vector in V can be written as a unique linear combination of the v_i's. The dimension of the space is defined to be the length of any basis, e.g. $\dim(V) = n$.

Definition 4 (Sum, Direct Sum)**.** For subspaces U, W of V, their *sum* $U + W = \{u + w : u \in U, w \in W\}$ is also a subspace of V. Note $\dim U + \dim W \geq \dim U + W$ except when $U \cap W = 0$ in which case we write the sum as $U \oplus W$ and call it a *direct sum*.

Definition 5 (Linear Map)**.** Let V, W be vector spaces over a field $\mathbb{F}$. A linear map $T : V \to W$ is a map that satisfies the following properties:

- For all $v, w \in V$, $T(v) + T(w) = T(v + w)$, and
- for all $\lambda \in \mathbb{F}, \lambda T(v) = T(\lambda v)$.

Definition 6 (Null Space and Range)**.** Let $T : V \to W$ be a linear map. The null space (also known as the kernel) of T is the set of vectors $\text{Null}(T) = \{v \in V : T(V) = 0\}$ and is a subspace of V. The range of T is the set of vectors $\text{Range}(T)\{w \in W :$ there exists $v \in V$ such that $T(v) = w\}$ and is a subspace of W.

Theorem 1 (Fundamental Theorem of Linear Maps)**.** Suppose V is finite dimensional and $T : V \to W$ is a linear map. Then $\dim(V) = \dim(\text{Null}(T)) + \dim(\text{Range}(T))$.

Definition 7 (Isomorphism Between Vector Spaces)**.** An isomorphism between two vector spaces is an invertible linear map, i.e. a Linear map $T : V \to W$ with some other linear map $S : W \to V$ such that $S \circ T = id_V$ and $T \circ S = id_W$. Moreover, two finite-dimensional vector spaces over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.

Theorem 2 (Characterization of Injectivity)**.** A linear map $T : V \to W$ is injective if and only if $\text{Null}(T) = \{0\}$.

Theorem 3. A linear map is invertible if and only if it is injective and surjective.

Definition 8 (Vector Space of Linear Maps)**.** Given any two vector spaces V and W over a field $\mathbb{F}$, the set of linear maps from V to W, written $\mathcal{L}(V, W)$, has the structure of a vector space over $\mathbb{F}$. Addition and multiplication are given point-wise $(aT + bS)(v) = aT(v) + bS(v)$ and we have the zero map $0(v) = 0$.

Definition 9 (Matrix of Linear Map)**.** Given two vector spaces V and W over $\mathbb{F}$ with chosen bases $\alpha = \{v_1, \ldots, v_n\}$ and $\beta = \{w_1, \ldots, w_m\}$ we can write

$$\mathcal{M}_{\alpha,\beta}(T) = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}$$

where the scalars $a_{i,j} \in \mathbb{F}$ are the unique numbers such that $Tv_j = a_{1,j}w_1 + \cdots a_{m,j}w_m$. The jth column of $\mathcal{M}(T)$ is the vector Tv_j written as a column vector in W under the basis β.

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Section II: Exercises

1. Suppose that U is a subspace of a finite-dimensional vector space V. Prove that there is a subspace W of V such that $U \oplus W = V$.
2. Let $T : P_3(\mathbb{R}) \to P_2(\mathbb{R})$ be the map $T(p) = p'' + p'$. For example $T(x^3 + 2) = 6x + 3x^2$. Compute the matrix $\mathcal{M}(T)$ with respect to the standard basis of $P_3(\mathbb{R})$ and $P_2(\mathbb{R})$.
3. Let V be the subspace of $\mathbb{P}_2$ of polynomials of degree less than or equal to 2 such that $p'(1) = p(2)$. Construct a basis of V and prove that it is indeed a basis.
4. Let $S : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be the map $S(p) = \int_0^x p(t)\,dt - p'(x)$. For example $S(x + 1) = \frac{1}{2}x^2 + x - 1$.
   i) Argue that S cannot be surjective.
   ii) Calculate the subspaces Null(S) and Range(S).
   iii) Calculate the matrix of S with respect to the standard bases of $\mathcal{P}_2(\mathbb{R})$ and $\mathcal{P}_3(\mathbb{R})$.
5. Let V and W be two finite-dimensional vector spaces over a field $\mathbb{F}$. Prove that if V and W are isomorphic, then they have the same dimension.
6. Does there exist a linear map $T : \mathbb{R}^4 \to \mathbb{R}^2$ such that
$$\text{Null}(T) = \{(x, x, x, x) \in \mathbb{R}^4 | x \in \mathbb{R}\}?$$
7. Prove that the intersection of subspaces is a subspace. What about union?
8. Suppose that V is a finite-dimensional vector space with $T \in \mathcal{L}(V)$, and fix a basis $v_1, \ldots, v_n$ of V. Prove T is an isomorphism if and only if $Tv_1, \ldots, Tv_n$ are linearly independent.
9. Let V be a finite dimensional vector space with subspaces $U, W \subset V$. Show that if $U \cap W = \{0\}$ then there exists an isomorphism $T : U \times W \to U \oplus W$.
10. Let $\mathbb{F}^\infty$ be the vector space of all sequences $(a_1, a_2, \ldots)$ with $a_j \in \mathbb{F}$.
    i) Show that the right shift map $R : \mathbb{F}^\infty \to \mathbb{F}^\infty$ given by $R(a_1, a_2, \ldots) = (0, a_1, a_2, \ldots)$ is injective but not surjective.
    ii) Explain why it is impossible for a linear map $T : V \to V$ from a finite dimensional space to itself to have this property.
11. Give an example of a map $T \in \mathcal{L}(\mathbb{R}^2)$ with $\mathbb{R}^2 = \ker T \oplus \text{range } T$.
12. Say $D \in \mathcal{L}(\mathcal{P}_3(\mathbb{R}), \mathcal{P}_2(\mathbb{R}))$ is the differentiation map $D(p(x)) = p'(x)$. Find bases α, β such that
$$\mathcal{M}_{\alpha,\beta}(D) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$
13. Let V be a real vector space, and let $P : V \to V$ be a linear map satisfying $P^2 = P$. Such a linear map is called a projection. Show that $V = \text{null } (P) \oplus \text{range } (P)$.
14. Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.
15. Let V be a vector space over $F = \mathbb{R}$ or $\mathbb{C}$. Let $v_1, ..., v_n \in V$ be a list of vectors in V. For each $i \in \{1, ..., n\}$ define
$$w_i := \textstyle\sum_{j=1}^{i} j v_j.$$
Prove that span($v_1, ...v_n$)=span($w_1, ...w_n$).

(7) Let U, V be subspaces of W. So, U and V are vector spaces. Consider $U \cap V$:

1. $0 \in U$, $0 \in V$ b/c v.s., so $0 \in U \cap V$
2. Let $u, v \in U \cap V$. So, $u, v \in U$ and $u, v \in V$. Thus, $u+v \in U$, $u+v \in V$. So, $u+v \in U \cap V$.
3. Let $u \in U \cap V$. So, $u \in U$, $u \in V$. Thus, $\lambda u, \lambda v \in U, V$ for $\lambda \in \mathbb{F}$. So, $\lambda u \in U \cap V$.

what about union? → Use counter: x,y axis

1. $0 \in U \cup V$ b/c it's in the union of two v.s. ($0 \in U, V$)
2. Let $u, v \in U \cup V$ No way to guarantee $u+v$ is in both U and V.

(A:) Let $U, V \subseteq W$. So, U, V are v.s.

1. $0 \in U$, $0 \in V$, so $0 \in U \cap V$.
2. Let $x, y \in U, V$. So, $x, y \in U$, and $x, y \in V$. Thus, $x+y \in U$ and $x+y \in V$. So, $x+y \in U+V$.
3. Let $x \in U \cap V$ and let $\lambda \in \mathbb{F}$. So, $x \in U$, and $x \in V$. Also, $\lambda x \in U$ and $\lambda x \in V$. So, $\lambda x \in U \cap V$.

Thus, $U \cap V$ is a subspace. □

(A:) Now, $U \cup V$ is not always a subspace. Consider $U, V \in \mathbb{R}^2$, where $U = \{(a, 0) \mid a \in \mathbb{R}\}$ and $V = \{(0, b) \mid b \in \mathbb{R}\}$.

$(1, 0) \in U \cup V$ b/c $(1, 0) \in U$; $(0, 1) \in U \cup V$ b/c $(0, 1) \in V$.

$(1, 0) + (0, 1) = (1, 1)$. $(1, 1) \notin U$ and $(1, 1) \notin V$, so $(1, 1) \notin U \cup V$.

Counterexample!

(3) $a(2)^2 + b(2) + c = 2a(1) + b$

$4a + 2b + c = 2a + b \Rightarrow 2a + b + c = 0$

$\Rightarrow a = -\frac{1}{2}b - \frac{1}{2}c$

$\left(-\frac{1}{2}b - \frac{1}{2}c\right)x^2 + bx + c$

$\Rightarrow -\frac{1}{2}bx^2 - \frac{1}{2}cx^2 + bx + c \rightarrow b\left(-\frac{1}{2}x^2 + x\right) + c\left(-\frac{1}{2}x^2 + 1\right)$

Basis: $\mathcal{B} = \left\{-\frac{1}{2}x^2 + x,\ -\frac{1}{2}x^2 + 1\right\}$

To prove it's a basis, check L.I. and span.

$-\frac{1}{2}x^2 + x$ and $-\frac{1}{2}x^2 + 1$ are L.I. because they are not scalar multiples of each other and are nonzero. Any set of 2 nonzero vectors that are not s.m. is L.I..

For span, consider $ax^2 + bx + c \in P_2$.

$-\frac{1}{2}bx^2 - \frac{1}{2}cx^2 + bx + c = ax^2 + bx + c$

$x^2\left(-\frac{1}{2}b - \frac{1}{2}c\right) = ax^2$

$a = -\frac{1}{2}b - \frac{1}{2}c \Rightarrow 2a + b + c = 0$ ← we found a, b, c to satisfy.

(A) $p(x) = ax^2 + bx + c$, $p(x) \in P_2$

$p'(x) = 2ax + b$

$a(2)^2 + b(2) + c = 2a(1) + b$

$4a + 2b + c = 2a + b$

$c = 2a + b - 4a - 2b$

$c = -2a - b$

$p(x) = ax^2 + bx - 2a - b$

$a(x^2 - 2) + b(x - 1)$

$\beta = \{x^2 - 2, x - 1\}$

These vectors are L.I. as they are not scalar multiples of each other and are nonzero. Any two nonzero vectors that are not s.m. are L.I.

β spans because we have shown that any arbitrary polynomial $p(x)$ can be written as a L.C. of the vectors in β.

(4)

$S(p+q) = \int_0^x (p+q)\,dt - (p+q)^1$

$= \int_0^x p\,dt + \int_0^x q\,dt - p^1 - q^1$ (laws of integration, scalar mult.)

$S(\lambda p) = \int_0^x \lambda p\,dt - (\lambda p)^1$

$= \lambda \int_0^x p\,dt - \lambda p^1$

$= \lambda\left(\int_0^x p\,dt - p^1\right)$ (laws of integration, scalar mult.)

i) Not surjective b/c the dimension of the domain ($\mathbb{R}^2$) is 3, so range T cannot be greater than 3. But the dim of the codomain ($\mathbb{R}^3$) is 4, so range T is less than the dim of codomain, so T is not surjective.

ii) for $\int_0^x p\,dt - p^1$. $\int_0^x p\,dt$ and p^1 always have different orders, so they will never cancel out to 0. We need a polynomial p s.t. $S(p)$ goes straight to 0.

Lo can't be a constant polynomial, deg 1, or deg 2.

Lo $S(0) = 0$ is the only way to get 0.

So, null $S = \{0\}$. dim null $S = 0$, thus dim range $S = 3$.

iii) The standard basis of $P_2(\mathbb{R})$ is $(1, x, x^2)$,

$S(1) = x - 0 = 0x^3 + 0x^2 + 1x - 0$ or $(0x^3+0x+1,\ 0x^2+1x+0,\ 1x^2+0x+0)$

$S(x) = \frac{x^2}{2} - 1 = 0x^3 + \frac{1}{2}x^2 + 0x - 1$

$S(x^2) = \frac{x^3}{3} - 2x = \frac{1}{3}x^3 + 0x^2 - 2x + 0$

$$M(S) = \begin{bmatrix} 0 & 0 & 1/3 \\ 0 & 1/2 & 0 \\ 1 & 0 & -2 \\ 0 & -1 & 0 \end{bmatrix}$$

x^2 x 1

$[a\ \ b\ \ c]$

00

4 x 3

row col

(A) i) same as blue work

ii) Find dim null S and use rank-nullity, as it is easier to find than dim range S.

Consider $S(ax^2+bx+c)=0$.

$S(ax^2+bx+c)=0 \Rightarrow a\,S(x^2)+b\,S(x)+c\,S(1)=0$

$= a\left(\frac{x^3}{3}-2x\right)+b\left(\frac{x^2}{2}-1\right)+c(x)=0$

Each term has a different degree, so the only way to have this polynomial equal 0 is if $a=b=c=0$. This creates the 0 vector. Thus, the only polynomial such that $S(p)=0$ is 0.
So, null $S=\{0\}$, thus dim null $S=0$. So, by rank-nullity, then dim range $S=3-0=3$.

↳ This also generates a basis - spans because we can replace 0 with any arbitrary polynomial and L.I. b/c they are different degrees.

iii) $M(B)=\begin{bmatrix} S(1) & S(x) & S(x^2) \\ \vdots & \vdots & \vdots \end{bmatrix}$

$S(1)=x-0=0x^3+0x^2+1x-0$

$S(x)=\frac{x^2}{2}-1=0x^3+\frac{1}{2}x^2+0x-1$

$S(x^2)=\frac{x^3}{3}-2x=\frac{1}{3}x^3+0x^2-2x+0$

$\Rightarrow M[S]=\begin{bmatrix} 0 & 0 & 1/3 \\ 0 & 1/2 & 0 \\ 1 & 0 & -2 \\ 0 & -1 & 0 \end{bmatrix}$